

Henry Phan

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SUMMARY OF QUALIFICATIONS

- Highly motivated 4th-year Computer Engineering student with strong engineering analysis and report-writing skills developed through several years of innovative academic projects.
- Hands-on experience in assembling, testing, and troubleshooting embedded systems and microcontrollers, with a strong foundation in system-level architecture.
- Familiar with interpreting, creating, and updating technical diagrams and layout drawings, with academic and project experience utilizing AutoCAD and Quartus II.

TECHNICAL SKILLS

Languages: Python, Javascript, SQL, Java, C, C++, MATLAB, HTML, CSS, VHDL, HCS12 Assembly

Frameworks & Libraries: NumPy, JavaFX, discord.py, SQLite

Software/Tools: AutoCAD, Microsoft Office (Excel, Word, PowerPoint), Git, GitHub, Linux, Power BI, Quartus II

Methodologies: Software Development Life Cycle (SDLC), UML Modelling, Object-Oriented Analysis, Boolean/Control Logic

EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering (B.Eng), Computer Engineering

2023 – Present

- **Relevant Courses:** Electronic Circuits I, Electronic Networks, Communication Systems, Signals and Systems I & II, Microprocessor Systems, Digital Systems, Principles of Engineering Economics
- Anticipated 2028

EXPERIENCE

Head Counter

April 2022 – September 2023

Sugarhill Desserts

- Proposed seasonal promotional ideas to the manager to attract customers.
- Debugged POS system errors to restore payment processing and sales reporting functionality.
- Assisted in training new staff members in a fast-paced environment.
- Performed inventory stock procedures to maintain inventory levels at all times.

PROJECTS

Embedded Navigation Robot

November 2025

- Programmed an HCS12 microcontroller in Assembly with a team to autonomously navigate a maze, detecting roadblocks and intersections.
- Implemented sensor-scanning, motor-control, and navigation logic to process photodetector and bumper inputs for real-time robot control.
- Utilized interrupts, hardware registers, state-machine logic, and subroutines to implement time-critical navigation routines.
- Led project management to meet strict deadlines and tracked test results to monitor progress.

Bookstore Application

April 2025

- Collaborated with a team to develop a bookstore application with a single-window GUI using Java and JavaFX.
- Applied concepts such as SDLC and the State Design Pattern to design an efficient customer membership system.
- Designed UML Use Case and Class diagrams to model the app's structure and functionality.

CPU Design

November 2024

- Designed and implemented a VHDL-based CPU on an Altera FPGA board to execute arithmetic and logic operations.
- Utilized block diagrams and code to integrate components such as an ALU, D flip-flops, 3-to-8 decoders, encoders, and SSEGs.
- Verified CPU functionality through waveform simulations and hardware testing across all implemented operations.